BEING LOCALLY OF FINITE TYPE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'and we win' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base-change stability and Zariski locality of local finite type. Morphisms, Lemmas morphisms-lemma-base-change-finite-type and morphisms-lemma-locally-finite-type-characterize, invoked at proof lines 5960-5963. Supplies the two preliminary halves of fpqc locality.
- Finite type descends along a faithfully flat ring map. Algebra, Lemma algebra-lemma-finite-type-descends, invoked at proof lines 5974-5978. Gives that R -> A is of finite type.
- The general principle for a property that is fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof lines 5979-5980. Concludes the locality statement.

Source reference. Original lemma 00QP: sources/stacks-chow/tags.txt, source lines 975–975. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T4: sources/stacks-chow/tags.txt, source lines 2361–2361. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is locally of finite type" is fpqc local on the base.*

Proof. Being locally of finite type is preserved under base change, see Morphisms, Lemma 01T4. Being locally of finite type is Zariski local on the base, see Morphisms, Lemma 01T2. Finally, let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is locally of finite type. Let $U \subset X$ be an affine open. Then $U' = S' \times_S U$ is affine and of finite type over S'. Write $S = \mathrm{Spec}(R)$, $S' = \mathrm{Spec}(R')$, $U = \mathrm{Spec}(A)$, and $U' = \mathrm{Spec}(A')$. We know that $R \to R'$ is faithfully flat, $A' = R' \otimes_R A$ and $R' \to A'$ is of finite type. We have to show that $R \to A$ is of finite type. This is the result of Algebra, Lemma 00QP. It follows that f is locally of finite type. Therefore Lemma 02KP applies and we win. □